

Fraud Detection

#	Bài toán cụ thể	Function / journey	Data signal	AI/DS/ML	Output	Được gì	Rủi ro / mất gì
1	Account takeover risk dựa trên IP/device/session	Login, session, sensitive action	IP mới, device mới, location, user-agent, session age, login velocity	ML/anomaly/rule score	risk score, step-up auth	Rất cụ thể, hợp security + fintech	Cần mô phỏng session data
2	Credential stuffing detection	Login	failed login count, nhiều account cùng IP/device, password spraying	DS + anomaly/rule	block/challenge IP/account	Học security analytics, rất thực tế	Synthetic data là chính
3	Suspicious post-login payment detection	Sau login thành công	login anomaly, payment amount, recipient mới, time-after-login	ML/rule hybrid	hold/review/block	Gắn account security với money movement	Scope phức tạp hơn login fraud
4	New device + high-value transfer risk	Transfer/wallet payment	device first seen, amount lớn, recipient mới, time bất thường	ML-lite/rule score	step-up/manual review	Rất gần app ngân hàng/ví thật	Cần define threshold/rule tốt

5	Impossible travel login risk	Login/session	login location A rồi B quá nhanh, IP geo, device	Rule + anomaly	suspicious login alert	Cụ thể, dễ demo	Geo-IP giả lập có thể không thật
6	Session hijacking anomaly	Active session	session duration, token reuse, IP change mid-session, device mismatch	Anomaly detection	revoke/re-auth session	Học security + risk scoring	Dữ liệu session phải mô phỏng kỹ
7	MFA bypass / OTP abuse detection	Login/OTP verification	OTP fail count, resend count, SIM/device change, time pattern	Rule + ML-lite	challenge/block/review	Rất thực tế mobile banking	Nhạy cảm, cần synthetic
8	Device fingerprint risk scoring	Device binding	device_id, OS, emulator/root signal, reuse across accounts	Anomaly/graph	device risk score	Hợp ví điện tử/app tài chính	Dữ liệu device synthetic
9	IP reputation / proxy/VPN risk scoring	Login/payment	IP ASN, proxy flag, country mismatch, many accounts per IP	Rule + anomaly	IP risk level	Cụ thể, dễ làm module	Nếu không có real IP intel thì giả lập

10	Bot-like account activity detection	App/session behavior	action frequency, timing regularity, navigation pattern	Anomaly detection	bot risk score	Học behavioral analytics	Cần simulate behavior logs
11	QR merchant payment anomaly detection	QR/VietQR/merchant payment	amount, merchant, time, user baseline, merchant category	ML anomaly/classification	allow/review/block	Bám QR/cashless payment VN	Không có real QR data
12	High-risk first transaction detection	First payment/transfer	first transaction amount, recipient/merchant, device age	ML/rule	review/step-up	Hẹp, rõ, dễ demo	Dễ rule-based
13	Velocity-based wallet cash-out fraud	Withdraw/cash-out	số lần rút, amount, recipient, time window, device	Rule + anomaly	hold/review	Cụ thể, gần ví điện tử	Cần mô phỏng flow cash-out
14	Rapid transfer chain detection	P2P transfer	tiền đi qua nhiều account nhanh, amount similar, graph path	Graph analytics	suspicious flow chain	Có graph angle mạnh	Khó hơn nếu chưa biết graph

15	Mule account detection	Account receiving/transferring money	many inbound/outbound, low balance retention, new account	Graph + anomaly	mule account risk	Rất fintech/risk thực tế	Dữ liệu labels khó
16	Circular money movement detection	Wallet/bank transfer graph	A→B→C→A, repeated cycles, similar amounts	Graph anomaly	suspicious ring/cycle	Hay, có chiều sâu	Graph synthetic cần thiết kể tốt
17	Smurfing / transaction splitting detection	Transfer/cash-out	nhiều giao dịch nhỏ gần threshold, time window, recipient group	Rule + anomaly	structuring risk alert	Gần AML/fraud monitoring	Nhạy cảm, cần framing học tập
18	New beneficiary risk scoring	Add recipient/beneficiary	recipient age, first transfer, account name mismatch, context	ML/rule	risk warning	Cụ thể, dễ scope	Phụ thuộc data giả lập
19	Beneficiary takeover suspicion	Transfer beneficiary management	beneficiary changed, new device, high transfer soon after	Rule + anomaly	review beneficiary action	Rất gần app banking	Cần journey data
20	APP scam warning trước khi user chuyển tiền	User-initiated transfer	recipient mới, amount lớn, message/note,	NLP + rule/ML	warning/challenge	Rất thực tế, khác fraud truyền thống	Scam labels khó, dễ overclaim

			user behavior, scam keywords				
21	Suspicious transfer note / message detection	Transfer input	payment note, chat/support text, suspicious terms	NLP classifier	scam/social-engineering signal	Hợp NLP, function hợp	Text synthetic có thể giả
22	First-time recipient + pressure pattern detection	Transfer confirmation	recipient mới, amount lớn, repeated edits, fast confirmation	Behavioral DS + rule	scam-risk warning	Gần APP scam prevention	Cần simulate UX behavior
23	Remote access scam signal detection	Transfer/session	unusual session duration, accessibility use, copy-paste, new payee	Rule/anomaly	possible remote scam alert	Có góc nhìn mới	Data khó, cần giả lập cẩn thận
24	Merchant fraud / fake merchant risk	Merchant onboarding/payment receiving	merchant mới, volume tăng bất thường, refund rate, many users same device	ML anomaly/graph	merchant risk score	Khác user fraud, hợp payment ecosystem	Data khó hơn
25	Merchant collusion detection	User–merchant transaction graph	same users paying same merchant,	Graph anomaly	collusion cluster	Rất hợp ví/super app	Graph khó, cần synthetic tốt

			cashback loop, refund loop				
26	Abnormal merchant volume spike detection	Merchant monitoring	payment count/amount by hour/day, historical baseline	Time-series anomaly	merchant spike alert	Cụ thể, dễ visualize	Có thể là business growth thật
27	Merchant refund abuse detection	Merchant refund behavior	refund rate, refund amount, repeated users, time-to-refund	ML/anomaly	merchant refund risk	Hợp merchant risk	Cần phân biệt legit refund
28	Fake merchant onboarding risk	Merchant registration	business info, owner reuse, bank account reuse, device/IP	Rule + graph/ML	onboarding risk score	Gần KYB/merchant acquiring	Lấn KYB, nhưng fraud-specific
29	Promotion/cashback abuse detection	Campaign, voucher, cashback	nhiều account/device/IP, repeated merchant, same pattern	Graph/ML/rule	suspicious users/groups	Rất hợp ví điện tử/super app	Cần mô phỏng campaign data
30	Referral fraud / fake account farming	Referral campaign	shared device/IP, same bank account, similar profile, referral graph	Graph analytics + anomaly	suspicious referral cluster	Hay, có graph angle	Khó hơn nếu chưa biết graph

31	Voucher farming detection	Voucher/campaign redemption	many accounts, same device, same merchant, redemption timing	Anomaly/graph	voucher abuse alert	Rất cụ thể với ví/super app	Data synthetic
32	Cashback loop detection	Cashback campaign	pay-refund-repeat, same merchant/user cluster, repeated amount	Graph + sequence anomaly	cashback loop cluster	Bài có chiều sâu	Cần thiết kế campaign flow
33	Multi-account abuse detection	Account creation + campaign usage	device reuse, IP reuse, phone/email pattern, redemption history	Entity resolution + anomaly	account cluster risk	Rất thực tế app consumer finance	Dễ trùng KYC duplicate, cần định vị abuse
34	Synthetic user cluster detection	Account farming	similar profile pattern, device/IP overlap, low normal usage	Clustering/graph	synthetic cluster	Hợp abuse analytics	Data khó validate
35	Refund abuse detection	Refund workflow	refund frequency, order/payment/refund ratio, merchant/user pattern	ML/rule	review refund	Cụ thể, thực tế e-commerce/fintech	Cần design dữ liệu

36	Partial refund abuse detection	Refund lifecycle	repeated partial refunds, cumulative refund, merchant/user link	Rule + anomaly	suspicious partial refund	Function hẹp, nhiều edge cases	Dễ sang ops nếu không có abuse angle
37	Chargeback/dispute risk prediction	Card/online payment	merchant, amount, user history, refund/dispute history	ML classification	chargeback risk	Học payment risk thật	Dataset khó, synthetic
38	Friendly fraud / first-party misuse detection	Dispute/chargeback	user paid then disputes, history, delivery/refund pattern	ML/rule	first-party misuse risk	Rất payment-risk specific	Nhạy cảm, labels khó
39	Promo-refund combined abuse	Campaign + refund	campaign redemption, refund after cashback, repeated pattern	Graph/rule/anomaly	abuse risk	Hợp ví/super app	Cần mô phỏng campaign + refund
40	Card testing / payment method testing detection	Checkout/payment method add	many small attempts, many cards/accounts, fail pattern	Anomaly/rule	card testing alert	Rất thực tế payment fraud	Nếu không làm card, có thể dùng “payment method testing”

41	Small-amount probing detection	Payment/transfer	nhiều giao dịch nhỏ, new recipient/merchant, fail/success mix	Anomaly detection	probing risk alert	Cụ thể, dễ demo	Dễ nhầm với user behavior thật
42	Abnormal failed-payment pattern as fraud signal	Checkout risk	repeated failures, many cards/accounts/IPs, velocity	ML/anomaly	fraud-risk alert	Phân biệt với Payment Ops bằng abuse intent	Cần không trộn với generic failure classifier
43	Stolen payment method use suspicion	Checkout/wallet top-up	new card, new device, high amount, billing mismatch giả lập	ML/rule	stolen-method risk	Hợp payment risk	Data synthetic
44	Wallet top-up then immediate cash-out risk	Wallet top-up/cash-out	top-up method, time-to-cashout, amount, device	Rule + anomaly/ML	laundering/abuse risk	Gần ví điện tử	Nhạy cảm, cần framing risk prototype
45	Top-up abuse / promo arbitrage detection	Wallet campaign/top-up	top-up source, cashback, withdrawal, repeated cycles	Graph/anomaly	arbitrage abuse cluster	Rất hợp ví/campaign	Phức tạp hơn

46	Account opening to first fraud event prediction	New account monitoring	KYC risk, device, first actions, campaign usage	ML survival/classification	early fraud risk	Hợp lifecycle modeling	Labels khó
47	Dormant account reactivation risk	Inactive account becomes active	long dormancy, new device, password reset, high transfer	ML/rule	reactivation risk	Rất cụ thể, thực tế	Cần account history
48	Compromised account behavior change detection	Existing account monitoring	spending/transfer pattern shift, device/IP change	Change-point/anomaly	behavior-change risk	Không cần fraud labels nhiều	False positives có thể cao
49	User-merchant graph anomaly detection	Transaction graph monitoring	bipartite graph user-merchant, edge weights, time	Graph ML/anomaly	suspicious subgraph	Rất hay cho DS/ML	Khó hơn, cần graph kiến thức
50	Fraud ring detection bằng shared entities	Cross-account risk	shared phone, device, IP, bank account, address	Graph/community detection	fraud ring candidates	Rất đáng học nếu thích graph	Synthetic data khó
51	Risk feature store for fraud model	Fraud scoring pipeline	velocity features, device features, graph features	DS/feature engineering	feature set/model input	Học MLE/data pipeline	Không phải problem nếu đứng riêng

52	Real-time transaction risk scoring	Payment authorization	transaction + user + device + velocity features	ML/rule hybrid	risk score + decision	End-to-end nhất cho fraud	Scope lớn, cần cắt nhỏ
53	Batch fraud monitoring for daily review	Daily ops review	previous day transactions, anomalies, clusters	DS/anomaly	review queue	Dễ làm hơn real-time	Ít production-real-time
54	Fraud alert prioritization	Analyst review queue	risk score, amount, customer value, rule hits	Ranking/ML	priority queue	Rất thực tế fraud ops	Cần giả lập analyst workload
55	Fraud alert explanation assistant	Analyst workflow	risk features, rules hit, transaction history	LLM structured explanation	analyst note + next action	Tốt nếu kết hợp model/rules	Không nên standalone
56	Fraud case similarity search	Analyst review	case features, past outcomes, notes	Embedding/vector search	similar past cases	Hợp AI search, hữu ích	Cần case history synthetic
57	False-positive reduction for fraud alerts	Fraud ops tuning	alert reason, reviewer outcome, user behavior	ML/ranking/threshold tuning	lower false-positive queue	Rất thực tế, có business value	Cần feedback labels

58	Fraud model threshold tuning	Model decision policy	scores, labels, cost of false positive/negative	DS evaluation	recommended threshold	Mentor DS để góp ý	Không nên làm standalone
59	Fraud feedback loop quality monitor	Analyst feedback	reviewer labels, disagreement, final outcome	DS analytics	feedback quality score	Học model improvement lifecycle	Cần mô phỏng ops feedback
60	Fraud model drift monitoring	Fraud model ops	feature distribution, score distribution, fraud rate	DS monitoring	drift alert	Governance add-on tốt	Nên làm add-on, không phải fraud core

Package 1 — Account Takeover & Post-login Payment Risk

Gồm:

- #1 Account takeover risk
- #3 Suspicious post-login payment
- #4 New device + high-value transfer
- #5 Impossible travel
- #7 MFA/OTP abuse
- #18 New beneficiary risk

Package 2 — QR / Wallet Transaction Risk Scoring

Gồm:

- #11 QR merchant payment anomaly
- #12 High-risk first transaction
- #13 Velocity-based cash-out
- #41 Small-amount probing
- #44 Top-up then immediate cash-out
- #52 Real-time transaction risk scoring

Package 3 — Promotion / Cashback / Referral Abuse Detection

Gồm:

- #29 Promotion/cashback abuse
- #30 Referral fraud
- #31 Voucher farming
- #32 Cashback loop
- #33 Multi-account abuse
- #39 Promo-refund combined abuse

Package 4 — Scam & Social Engineering Warning

Gồm:

- #20 APP scam warning
- #21 Suspicious transfer note
- #22 First-time recipient + pressure pattern
- #23 Remote access scam signal
- #55 Fraud alert explanation assistant

Package 5 — Merchant / Mule / Fraud Ring Detection

Gồm:

#15 Mule account detection

#16 Circular money movement

#24 Merchant fraud/fake merchant

#25 Merchant collusion

#49 User-merchant graph anomaly

#50 Fraud ring detection